METRO TICKET GENERATING SYSTEM

IN

SERVICENOW

Complete Project Documentation

ServiceNow – Service Catalog, Flow Designer, Fare Automation,

QRCode Generation and Notifications

Project	**Metro Ticket Generation System**
Platform	ServiceNow
Primary Interface	Service Portal
Core Features	Station Selection → Fare Calculation →Validation →Ticket Generation → QR Code
Automation	Catalog Client Scripts, UI Policies, Process Automation Engine, Notifications

Project Completed By:

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Project Overview

The Metro Ticket Generating System is a digital solution built on ServiceNow that simplifies and automates metro ticket booking. Passengers choose their source and destination stations, enter journey details, and receive a digital ticket with a unique QR code. The system calculates fares automatically and handles ticket processing using Service Catalog, Catalog Client Scripts, UI Policies, Flow Designer, Notifications, and Access Controls (ACLs).

The project is carried out in the following five phases:

Phase 1– Requirement Analysis and Planning: Defines business objectives, functional scope, stakeholders, and the execution of roadmap.

Phase 2– Backend Development and Configurations: Covers data architecture, automation logic, workflows, and business rules.

Phase 3– UI/UX Development and Customization: Focuses on interface design, navigation flow, and overall usability.

Phase 4– Data Migration, Testing and Security: Includes data handling, access control, QA testing, and data integrity validation.

Phase 5– Deployment, Documentation and Final Presentation: Covers deployment, troubleshooting, documentation, demonstration, innovation, scalability, and future enhancements.

The Project aims to provide a fast, secure, user-friendly, and paperless metro ticketing experience, and to show how ServiceNow can be used in a real-world transportation scenario.

Phase 1– Requirement Analysis and Planning

1.1 Business Objectives

The Main objective of Metro Ticket Generating System is to Digitize metro ticket booking using ServiceNow for fast, accessible, and automated ticket generation via QR codes.

- **Enhance Commuter Convenience**

Implement user-friendly digital ticketing options such as mobile apps, QR codes to reduce wait times, and streamline the ticketing process.

- **Increase Operational Efficiency**

Adopt Automated Fare Collection (AFC) systems to minimize manual interventions, reduce errors, and lower operational costs.

- **Promote Digital Payment Adoption**

Encourage the use of digital payment methods like UPI, credit/debit cards, and mobile wallets to facilitate cashless transactions and improve revenue collection.

- **Facilitate Data-Driven Decision Making**

Utilize data analytics from digital transactions to gain insights into passenger behavior, peak usage times, and route optimization opportunities.

- **Support Environmental Sustainability**

Reduce paper usage by transitioning to digital tickets, contributing to eco-friendly practices, and aligning with sustainability goals.

1.2 Functional scope

The system includes the following key functional features:

- Service Catalog items for ticket booking with variables like Source, Destination, Passenger Type, Number of Tickets.
- Automated fare calculation, QR code generation of digital tickets.

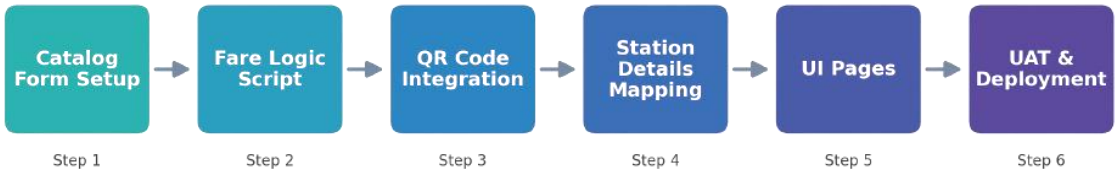
1.3 Stakeholder Mapping

The key stakeholders involved in the Metro Ticket Generating System are:

- Passengers (End Users): Book tickets, provide travel details, and access digital QR Tickets.
- Station Managers: Monitor ticket-related activities and assist passengers when required.
- Metro Operations Team: Manage ticketing operations, fares, routes, and overall system processes.
- IT Administrators: Configure, maintain, secure, and troubleshoot the ServiceNow application.

1.4 Execution Roadmap

- Catalog Form Setup: Create the *Book A Metro Ticket* form with the required variables.
- Fare Logic Script: Calculate the fare automatically from the stations, journey type, and number of passengers.
- QR Code Integration: Generate a unique QR code for each digital ticket.
- Station Details Mapping**:** Store station details and map source to destination.
- UI Pages: Build a simple, user-friendly booking interface.
- UAT & Deployment: Test with users, fix issues, and deploy the system.



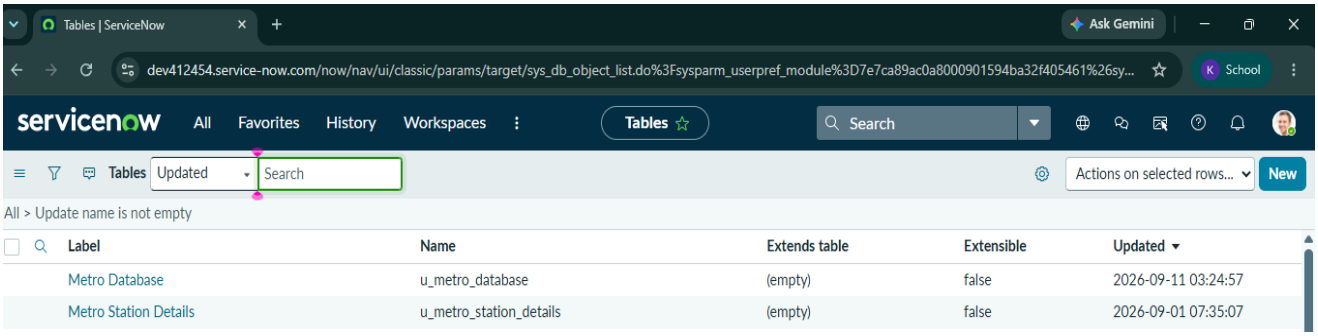
Phase 2-Backend Development and Configurations

2.1 Data Architecture

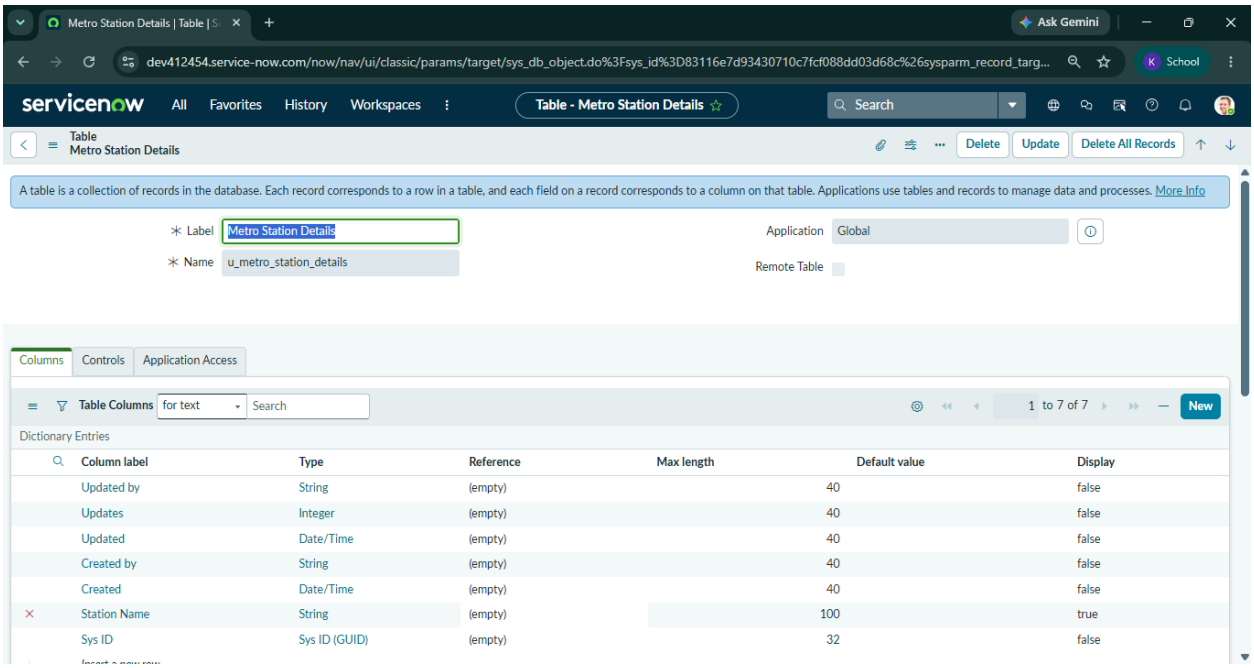
The data architecture defines how data is structured, stored, managed, and used within the Metro Ticket Generating System. In ServiceNow, a dedicated Metro Station Details table is created to maintain important station information required for ticket booking, station mapping, and fare calculation.

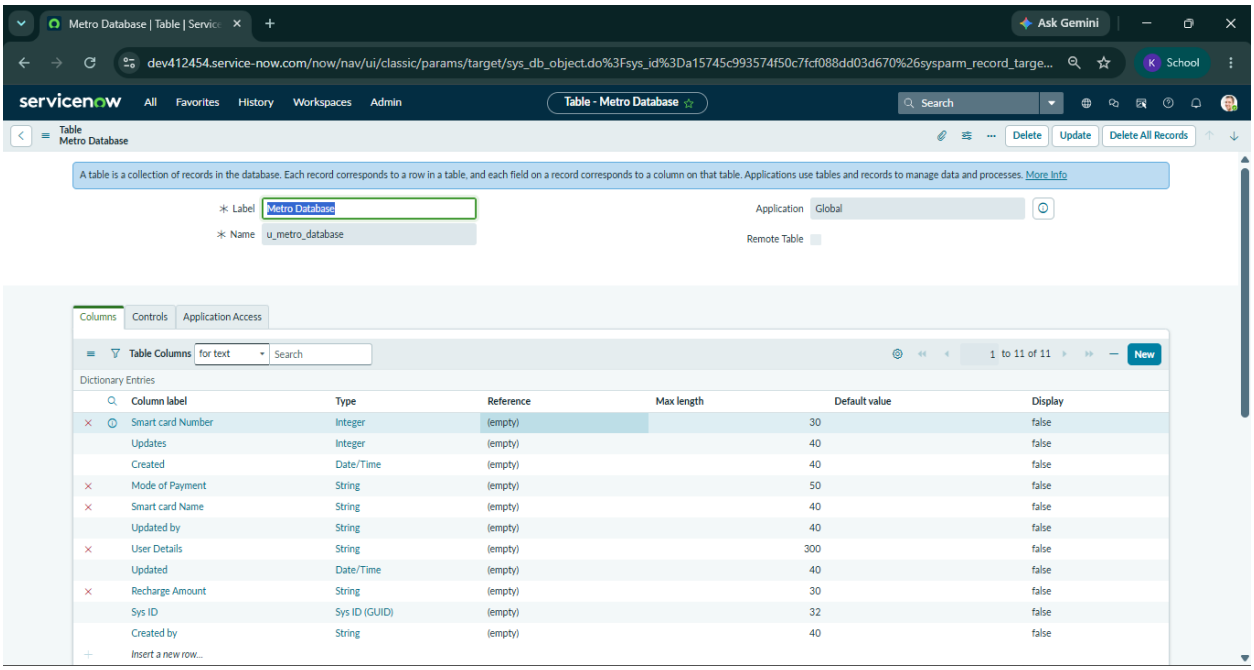
The implementation is carried out through the following activities:

- ✓ Creation of Table: A new Metro Station Details table is created under System Definition → Tables to store station-related information.

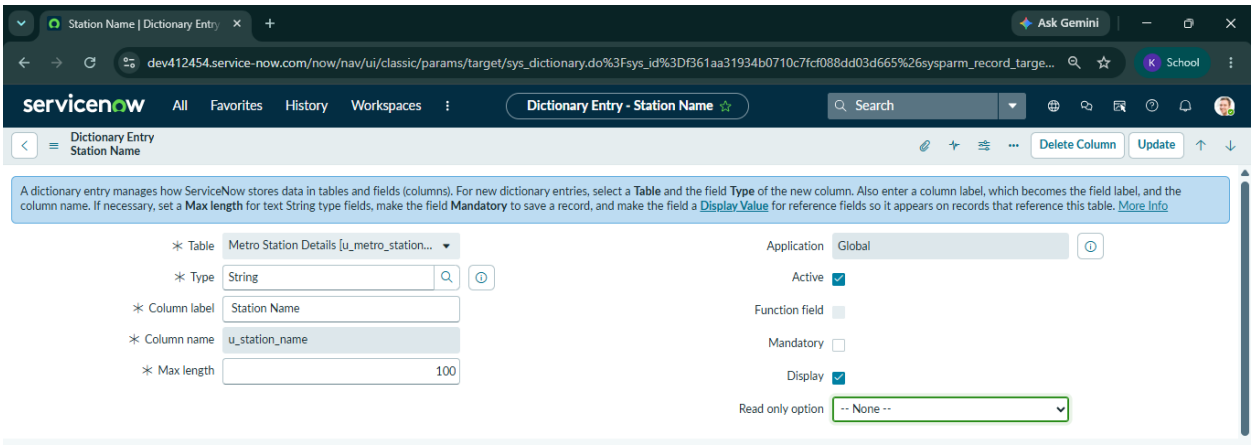


✓ Creation of Fields: Required fields are added to the table, such as Station Name.

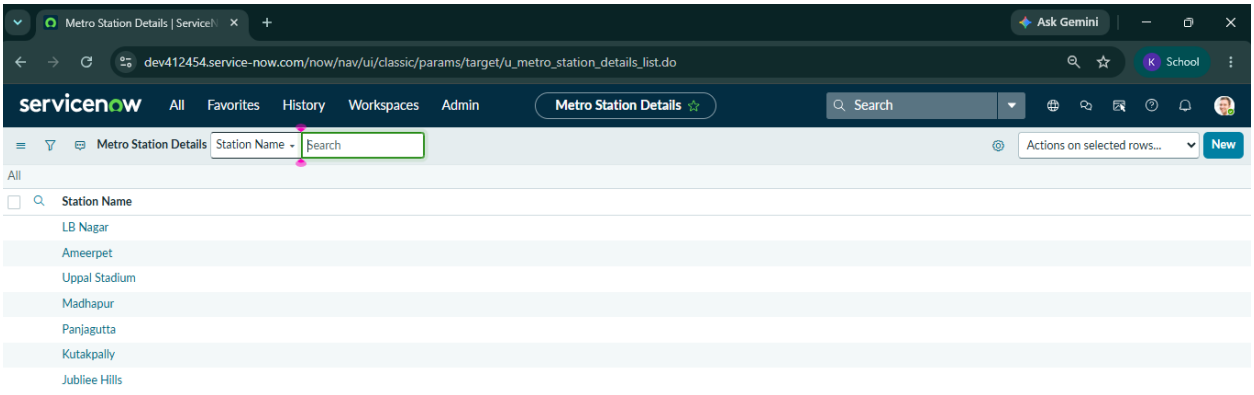




✓ Define Field Properties: Each field is configured with an appropriate data type such as String, Choice, Integer, Reference, or Boolean. Mandatory and read-only properties are applied wherever required.



✓ Form Configuration: The created fields are added to the corresponding form using the Form Designer, allowing authorized users to view and manage station information easily.



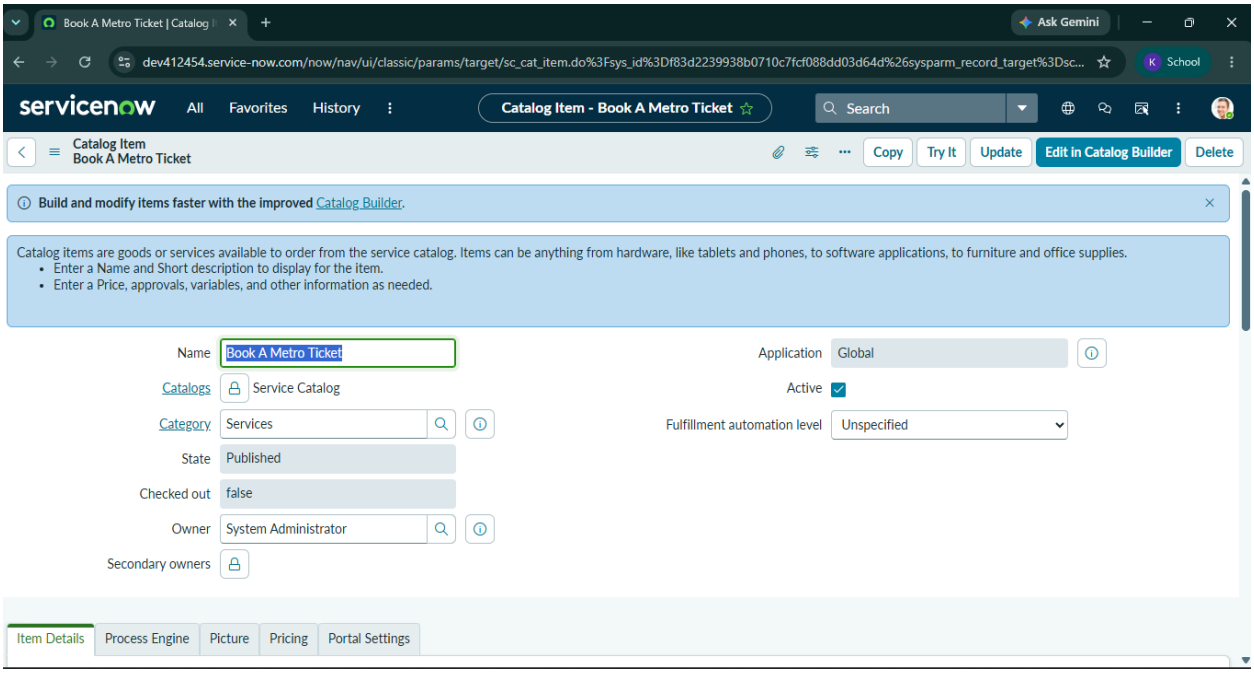
Phase 3– UI/UX Development and Customization

3.1 Interface Design

The Interface Design focuses on creating a simple, clear, and user-friendly interface for the Metro Ticket Generating System. A Service Catalog Item is created in ServiceNow to provide passengers with an easy way to enter their journey and ticket details.

➢ Service Catalog Creation

A catalog item named Book A Metro Ticket is created under the Service Catalog. This acts as the main interface through which passengers submit their ticket booking requests.



➢ Variables Configuration

Different variables are configured based on the information required for ticket booking. These variables allow users to enter or select details such as:

1. Source Station

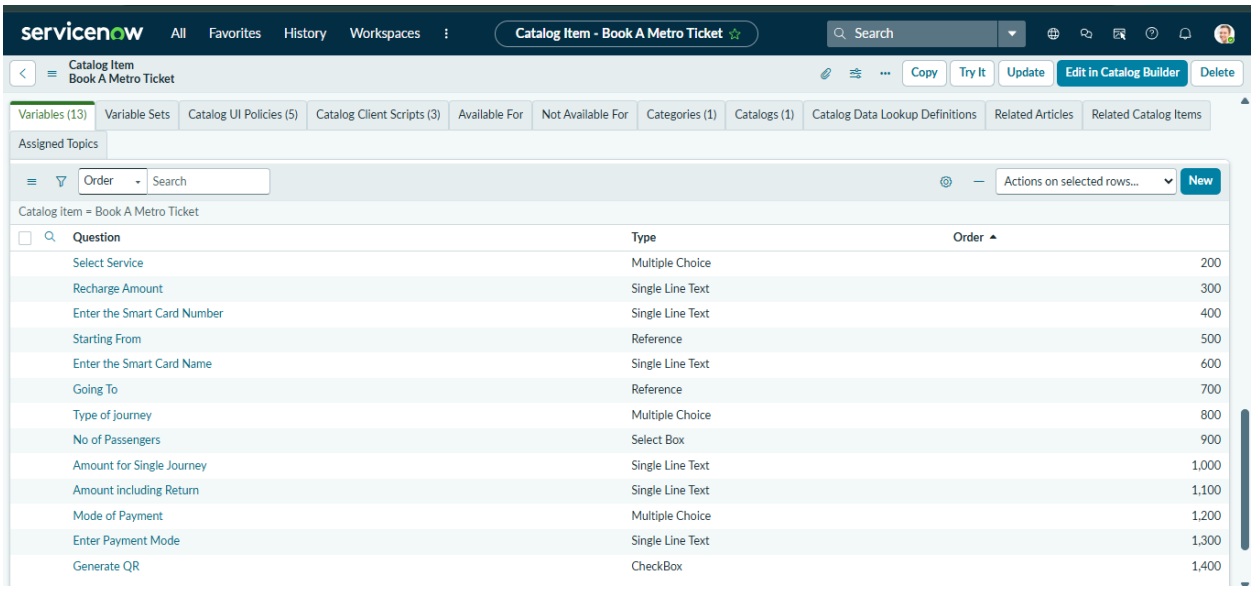
2. Destination Station

3. Passenger Type

4. Number of Tickets

5. Type of Journey

6. Mode of Payment



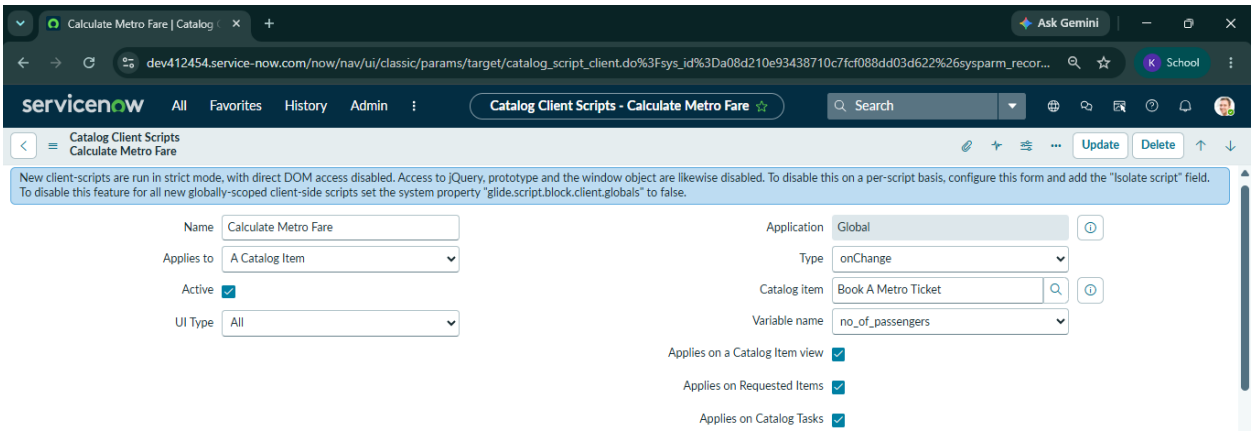
Fields can be configured as Choice, String, Reference, or Multi-line Text and can be made mandatory or read-only based on the requirement.

➢ Automation Logic

The Automation Logic is implemented using Catalog Client Scripts and Flow Designer to automate the metro ticket booking and fare calculation process. This reduces manual processing and ensures that the ticket information is generated accurately based on the user's selections.

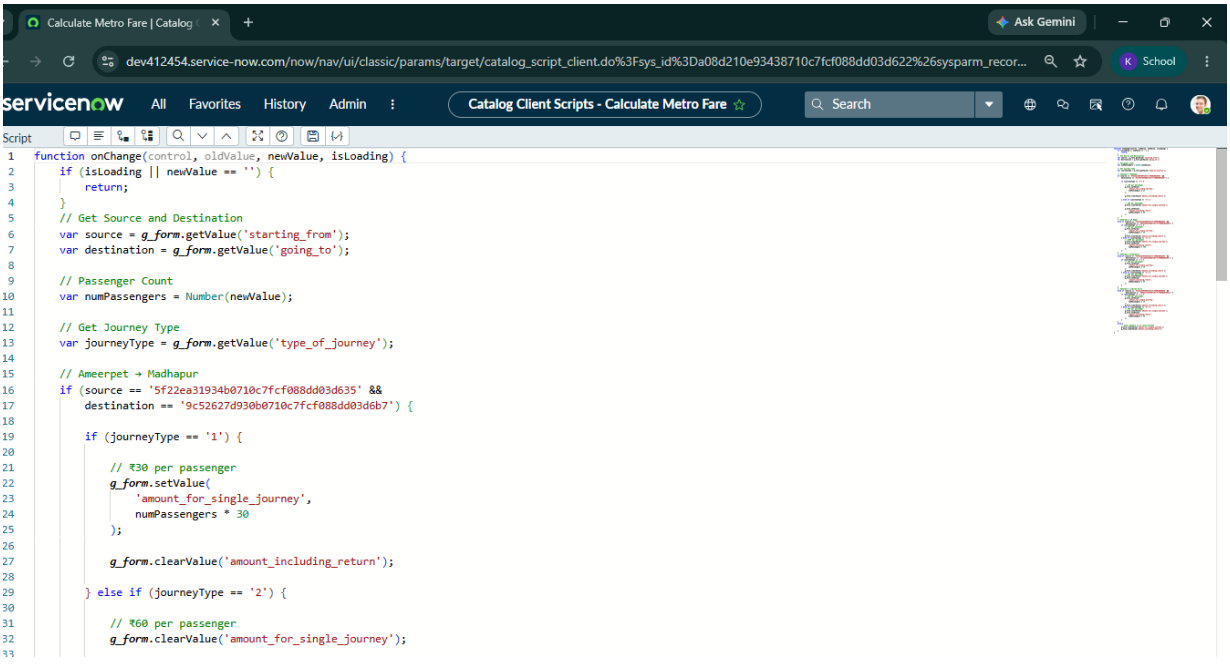
➢ Catalog Client Script– Automated Fare Calculation

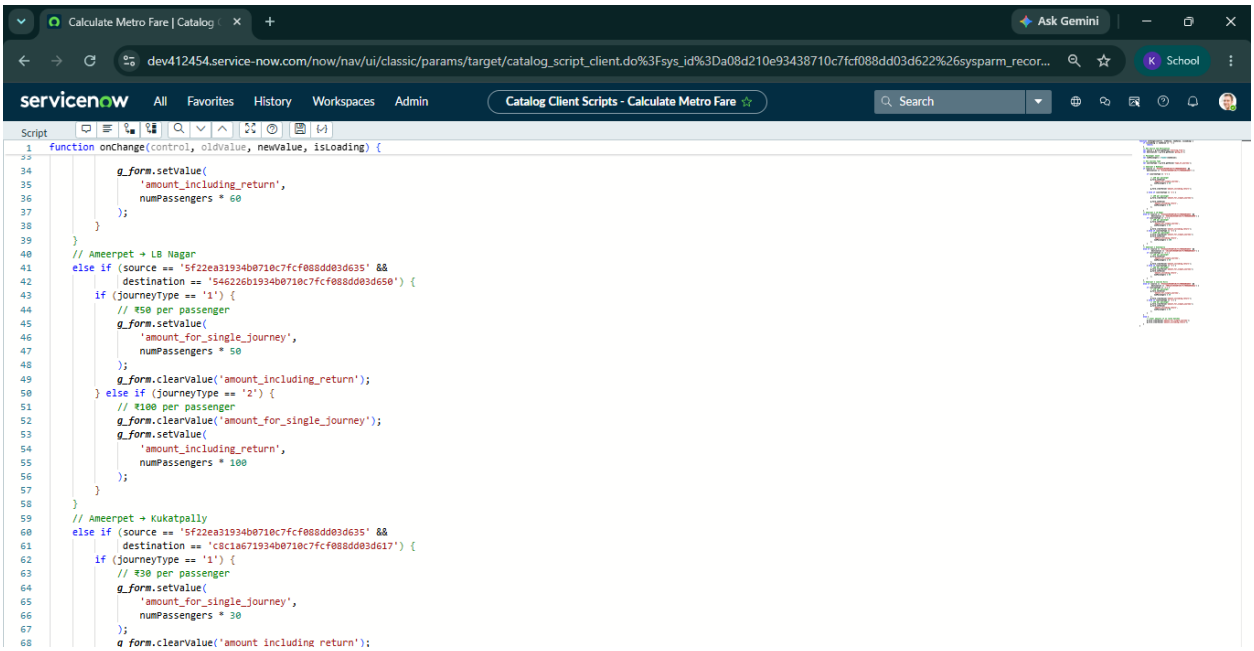
A Catalog Client Script is configured on the Book A Metro Ticket catalog item. The script runs when the user selects the Type of Journey and automatically calculates the fare based on the selected source, destination, journey type, and number of passengers.

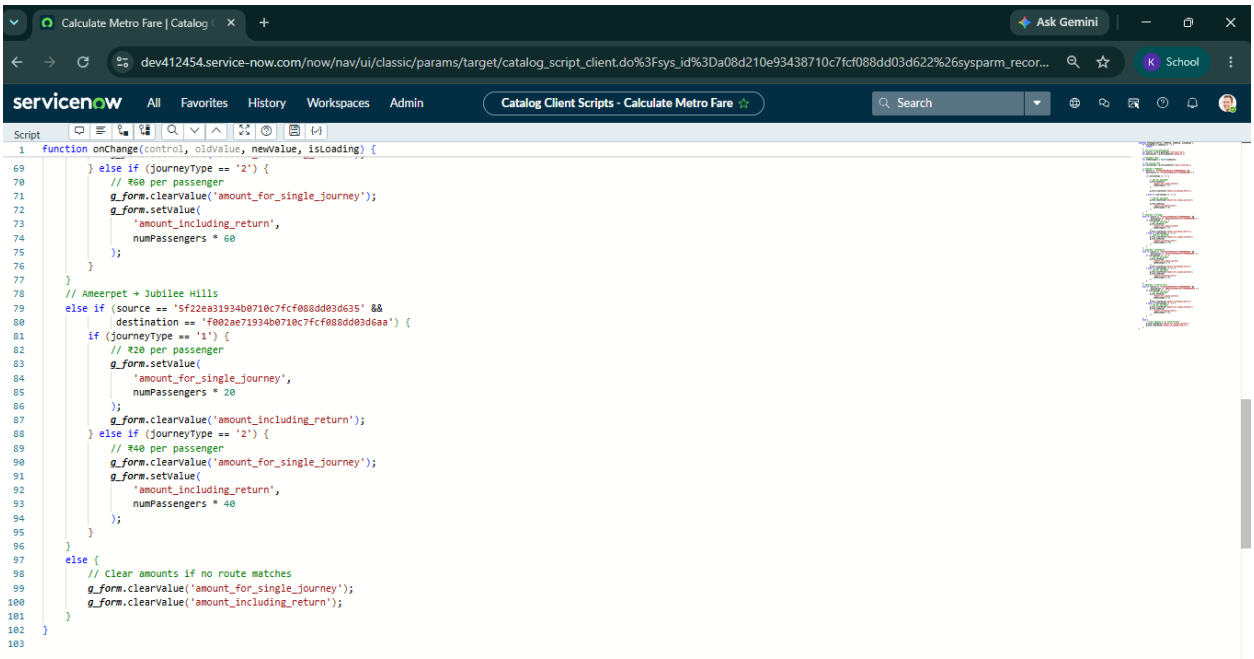


The script supports:

- ❖ Automatic fare calculation for different source and destination combinations.
- ❖ Separate calculations for single and return journeys.
- ❖ Automatic multiplication of fares based on the number of passengers.
- ❖ Clearing of irrelevant fare fields when the journey type changes.
- ❖ This provides dynamic fare calculation without requiring manual input from the user.



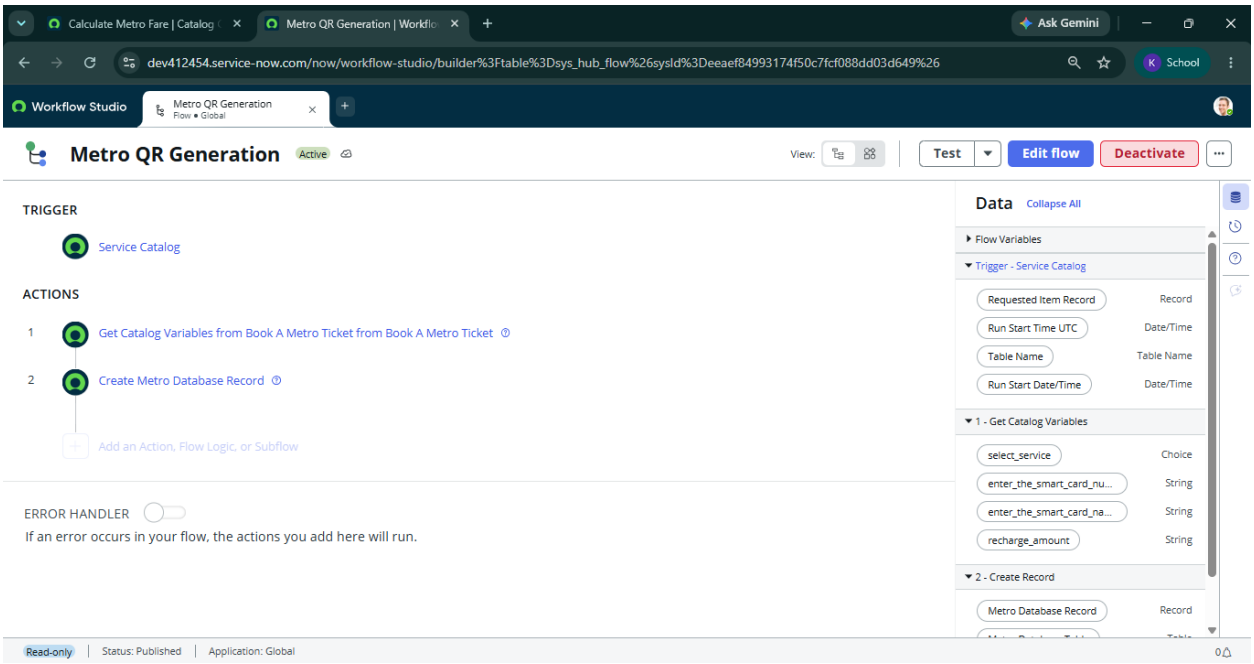




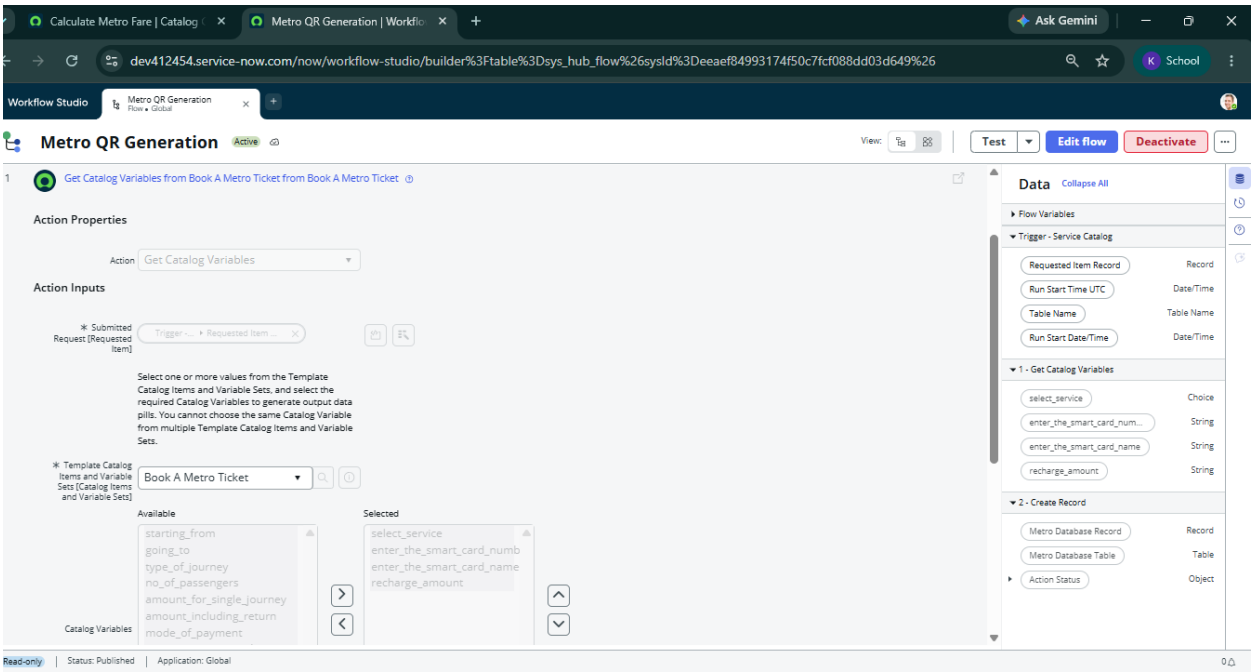
Flow Designer– Automated Ticket Processing

A Flow Designer flow named Metro Project is created to automate backend ticket processing. The flow performs the following steps:

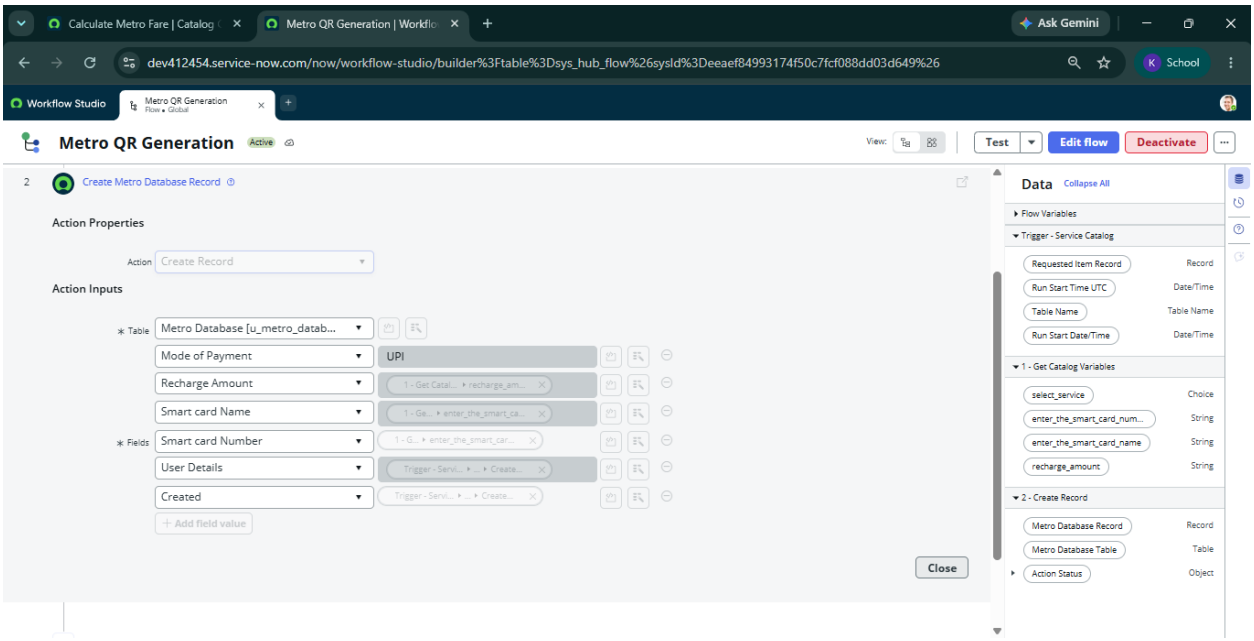
- ❖ Service Catalog Trigger: The flow starts when a metro ticket request is submitted.



- ❖ Get Catalog Variables: Retrieves the required variables from the Book A Metro Ticket catalog item.



- ➢ Create Record: Creates a new record in the Metro Database table using the submitted ticket information.

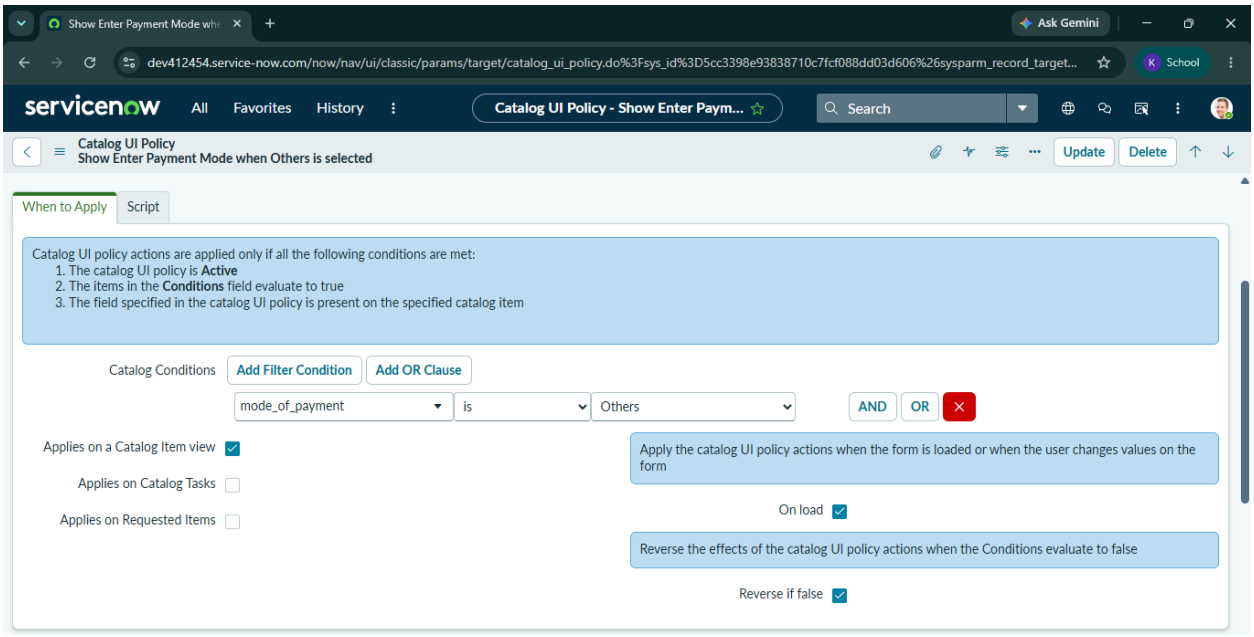


- Automated Processing: The flow ensures that the booking information is transferred and stored systematically for further ticket generation and processing.

This automation provides a smooth connection between the Service Catalog, fare calculation, and backend database, improving efficiency and reducing manual intervention.

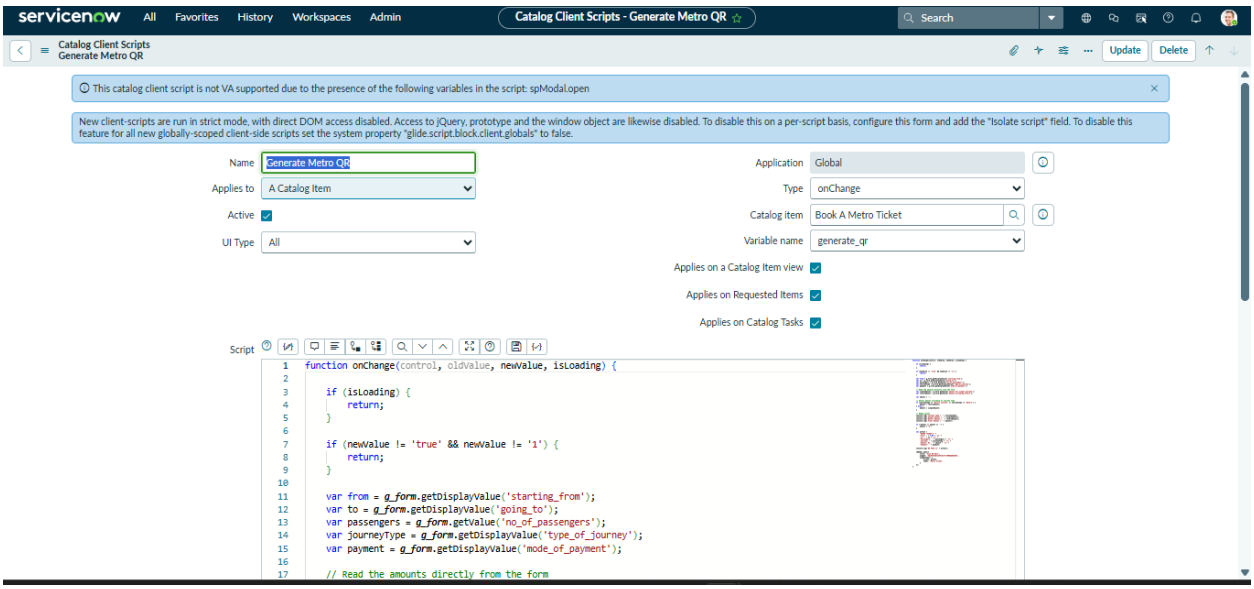
Catalog UI Policy– Conditional Fields

A Catalog UI Policy is configured to dynamically display the Enter Payment Mode field when the user selects Others as the Mode of Payment. This improves usability by showing only the fields relevant to the user's selection.



Catalog Client Script– QR Code Generation

An On Submit Catalog Client Script is implemented to generate a QR code when the ticket request is submitted. The script creates a unique ticket URL using the request identifier and displays the QR code through a modal window. This enables the passengers to access their digital tickets conveniently.

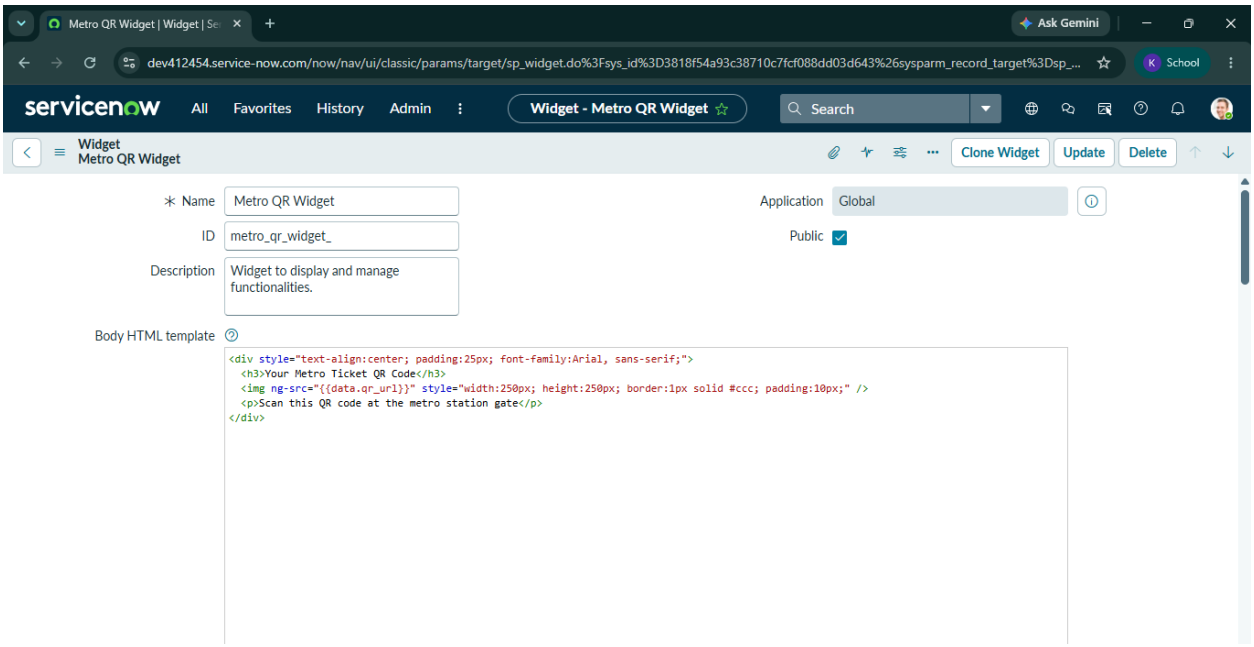


3.2 Widgets

The Metro QR Widget is a custom ServiceNow Service Portal widget developed to generate a digital QR ticket for the booked metro journey.

➢ Creation of Widgets in the Portal

Create Metro QR Widget: Creates a custom widget to generate and display the metro ticket details as a QR code using the submitted ticket information.

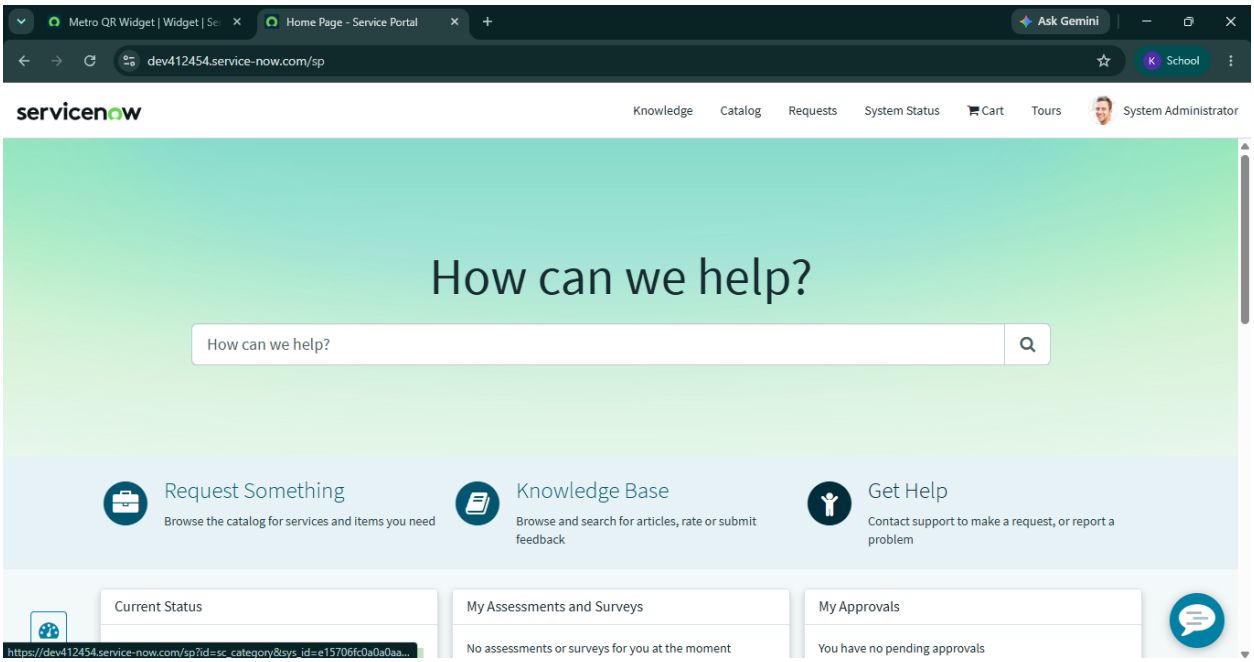


3.3 Navigation Flow

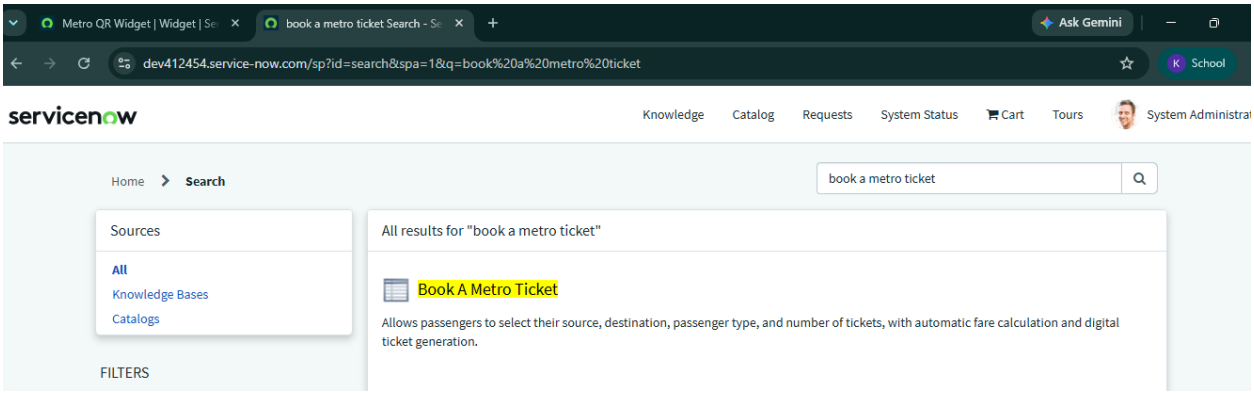
The navigation flow defines how users' access and completes the metro ticket booking process in ServiceNow. The process is designed to provide a simple flow from opening the service portal to submitting a ticket request.

➢ Navigation Process

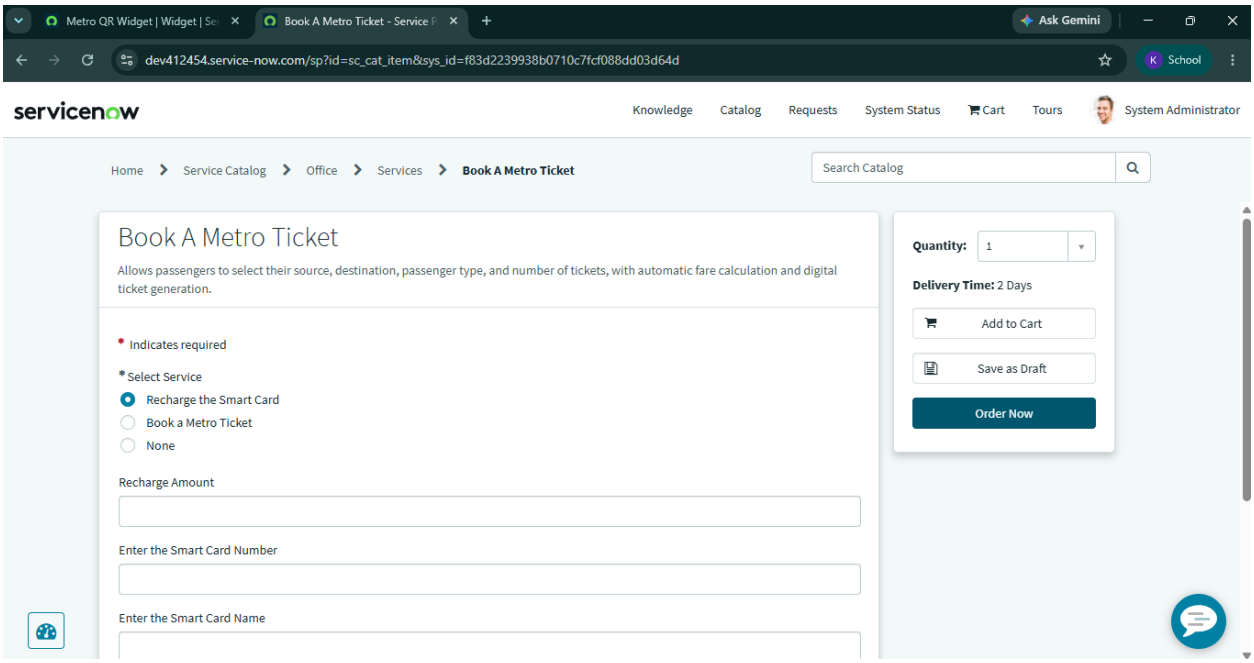
Access Service Portal: Log in to the ServiceNow instance and open the Service Portal.



Search Catalog Item: Search for Book A Metro Ticket in the Service Catalog



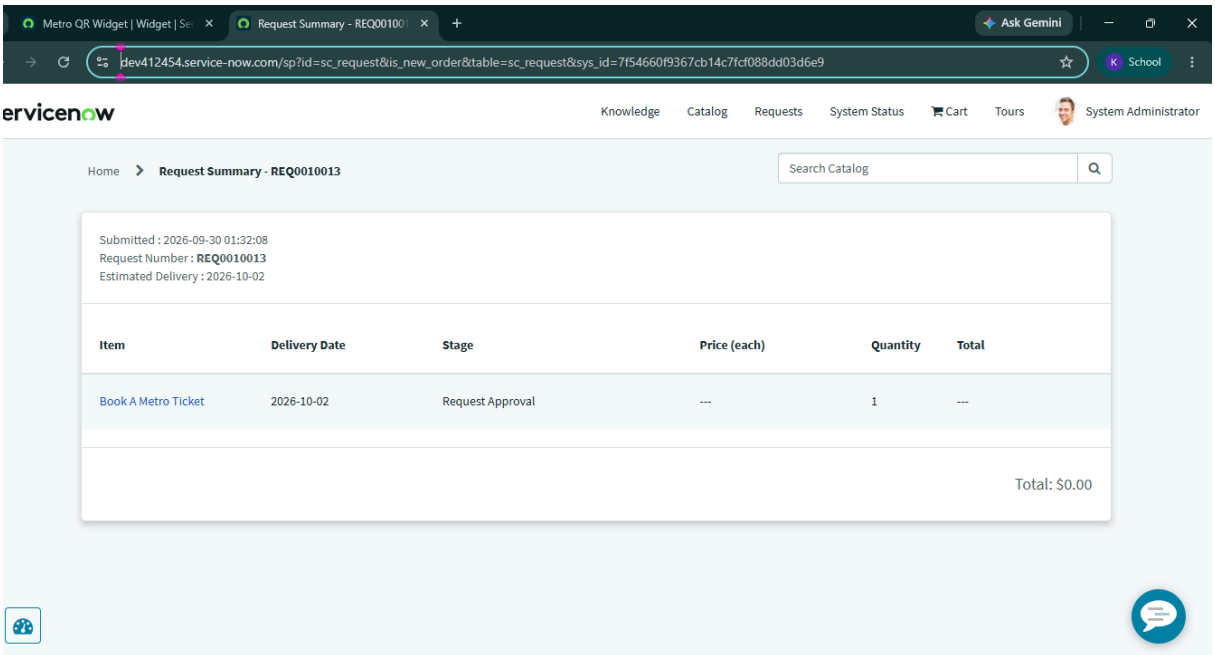
Enter Details: Fill in the required journey, passenger, and payment details.



Submit Request: Click Submit to create the metro ticket request.

Request Generation: ServiceNow generates a unique Request Number for the submitted booking**.**

Requested Item: The submitted ticket details can be viewed under the corresponding Requested Item (RTIM).



Usability

The Metro Ticket Generating System is designed with a strong focus on usability, ensuring that users can book tickets quickly and efficiently through a simple and intuitive interface. The catalog form provides clear labels, dropdown selections, and automated field populations to minimize user effort and reduce input errors. A key usability feature is the real-time fare preview, where the ticket amount is automatically calculated and displayed based on the selected source station, destination station, journey type, and number of passengers.

This navigation flow provides a clear and straightforward booking experience, allowing users to complete their ticket request with minimal steps.

Additional usability enhancements include:

a) Simple and user-friendly Service Portal interface.

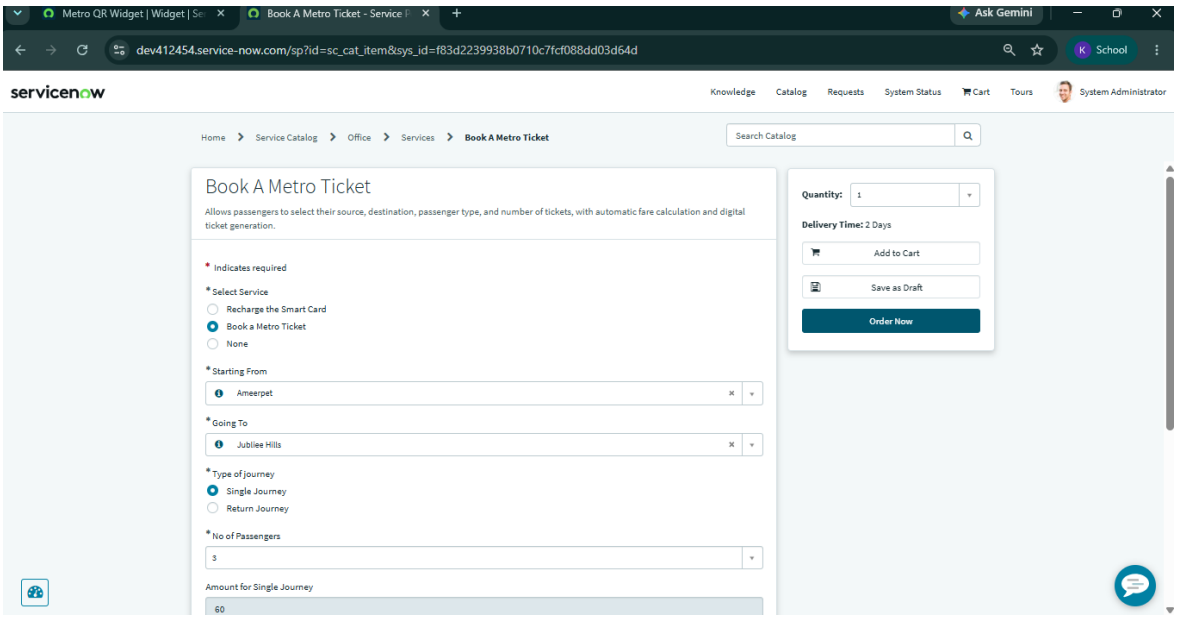
b) Dropdown-based station selection to reduce manual entry.

c) Automatic fare calculation without user intervention.

d) Dynamic field visibility based on user selections.

e) QR-based digital ticket generation for convenient access and travel.

These features provide a smooth and efficient booking experience while reducing the possibility of incorrect submissions.



Phase 4– Data Migration, Testing and Security

4.1 Data Handling

The Data Handling process ensures that ticket booking information submitted through the Service Catalog is securely captured and stored in the custom Metro Database table. The system uses Process Automation through Flow Designer to transfer and map catalog variables into structured database records.

The process includes:

1. Capture Catalog Data: User-entered ticket details are captured from the Book A Metro Ticket catalog item.

2. Get Catalog Variables: The Flow Designer Get Catalog Variables action retrieves the submitted variables.

3. Create/Update Record: A record is created or updated in the Metro Database table.

4. Variable Mapping: The Add Fields (+) option is used to map each catalog variable to its corresponding field in the Metro Database.

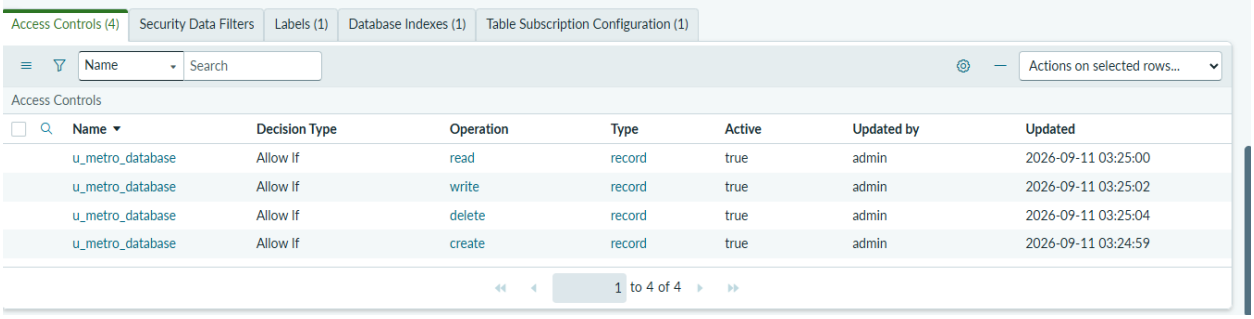
5. Structured Storage: The mapped information is stored systematically for ticket tracking, processing, and future reference. This automated approach ensures accurate, consistent, and efficient handling of ticket data while reducing manual data entry.

4.2 Access Control

Role-Based Access Control (RBAC) is implemented using Access Control Lists (ACLs) to protect the Metro Database and restrict unauthorized access to stored ticket information.

- Passenger/User Access: Users can directly book tickets and recharge their metro cards through the Service Catalog but cannot directly access the Metro Database table.
- Database Protection: The Metro Database is protected using ServiceNow ACLs to prevent unauthorized viewing or modification of records.

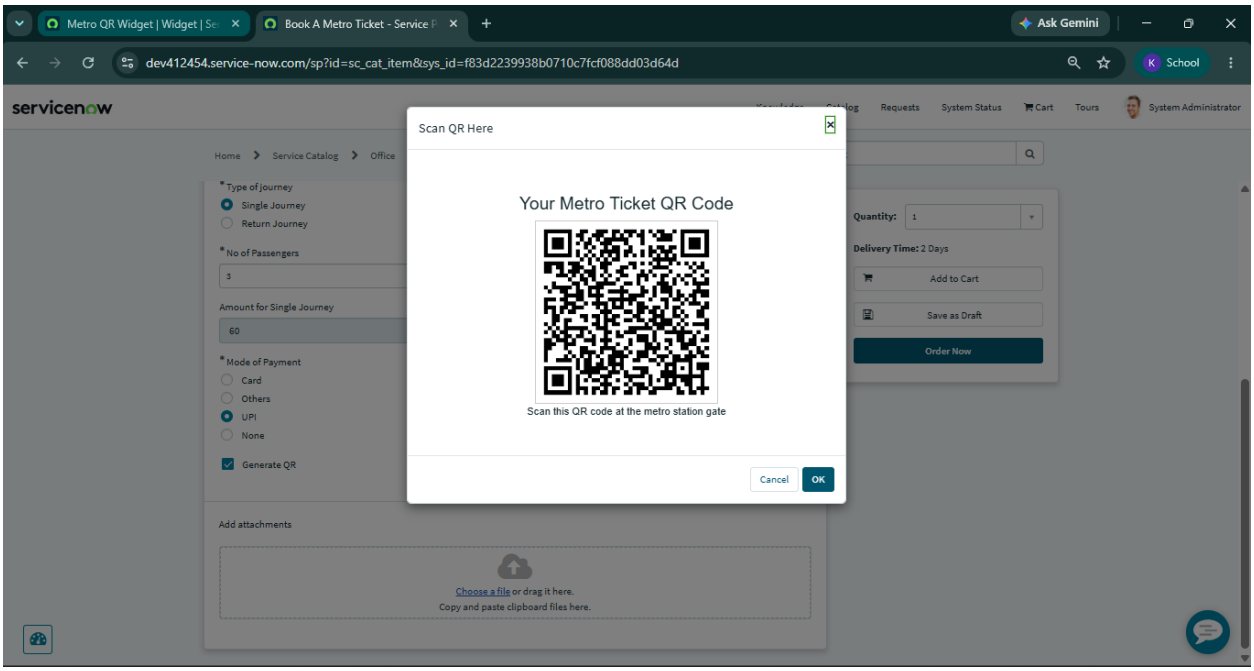
- Default ACLs: When the custom table is created, ServiceNow automatically generates the default table-level ACLs. The project uses these default ACL configurations.
- Role-Based Permissions: Access to sensitive data and database operations is controlled based on the user's assigned role.

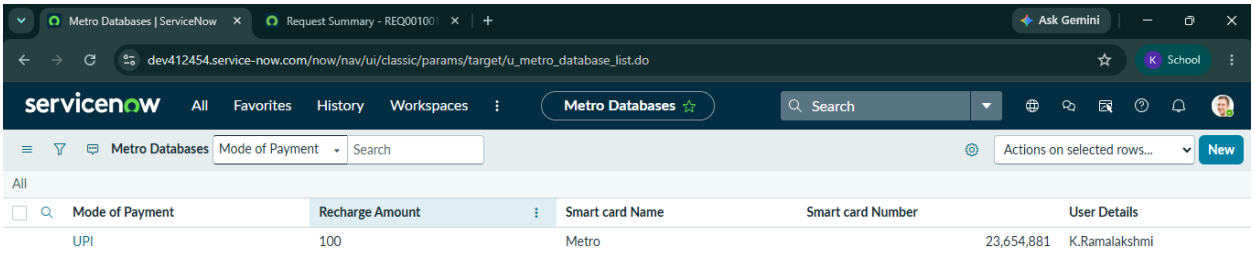


This approach ensures that passengers can use the required services without having direct access to backend database records, improving data security, and integrity.

4.3 QA Testing– Catalog Form, QR Generation and Visibility

QA testing was done to check whether the **Metro Ticket Generating System** works correctly. The ticket form was tested for required fields, station selection, passenger details, fare calculation, and field visibility. The QR code was also tested to make sure it is generated correctly with the ticket details. These tests helped confirm that the ticket booking process works properly.





3.4 Data Integrity

➢ Field Validation and Automation

Field validation was added to ensure users enter correct and complete booking details.

- ❖ **Mandatory Fields:** Required fields are validated to prevent incomplete submissions.
- ❖ **Auto-Populate:** Default values are automatically set using Client Scripts and UI Policies.
- ❖ **Field Validation:** Source, destination, fare, and travel date are checked for valid values.
- ❖ **Field Mapping:** Required information is automatically mapped and populated to maintain accurate data.

Phase 5– Deployment, Documentation and Final Presentation

5.1 Troubleshooting

During the deployment and testing stage, identified issues were analyzed and resolved to ensure smooth functioning of the Metro Ticket Generating System. Issues related to form visibility, fare calculation, field validation, QR code generation, and automation were tested and corrected. Troubleshooting helped improve the reliability and consistency of the application.

5.2 Adherence to Timelines

The project was developed according to a planned execution roadmap covering requirement analysis, backend configuration, UI development, testing, and final deployment. Each phase was completed systematically to ensure that the major functionalities were implemented within the planned schedule.

5.3 Innovation

The project introduces a digital approach to metro ticket booking using ServiceNow Service Catalog, automated fare calculation, and QR-based digital ticket generation. The combination of automated processing and QR tickets reduces manual effort and provides users with a faster and more convenient ticketing experience.

5.4 Functional Overview

The system allows users to select source and destination stations, enter passenger and journey details, calculate the applicable fare, submit the booking, and receive a digital QR-based ticket. ServiceNow components such as Service Catalog, Client Scripts, UI Policies, Process Automation, and Notifications work together to automate the complete ticket booking process.

5.5 Technical Blueprint

The technical design consists of the ServiceNow Service Catalog as the user interface, catalog variables for collecting travel details, scripts and UI policies for form behavior, backend data structures for storing information, automation for processing requests, and QR generation for digital ticket delivery.

5.6 Setup Manual

A setup manual was prepared to describe the steps required to configure and run the application. It includes ServiceNow instance setup, catalog item configuration, variable creation, client scripts, UI policies, data configuration, automation, QR generation, and testing procedures.

5.7 Project Demo Video Planning

The project demonstration video is planned to present the complete user journey. The demonstration begins with opening the metro ticket catalog item, entering travel details, viewing the calculated fare, submitting the request, and displaying the generated QR ticket.

5.8 Visual Demonstration

The visual demonstration showcases the actual working system through screenshots of the catalog form, station selection, fare calculation, conditional fields, ticket submission, and QR code generation. This provides clear evidence of the implemented functionalities.

5.9 Explanation Clarity

The project functionalities were presented in a clear and systematic manner. The demonstration explains the purpose of the system, booking workflow, automated

fare calculation, field validation, QR generation, and the role of ServiceNow components.

5.10 Realism and Quality

The system was designed to represent a realistic metro ticket booking environment. Realistic station selection, passenger information, journey types, fare calculation, digital ticket generation, and user interactions were incorporated to make the application practical and easy to understand. Screenshot: Final polished catalog form and QR ticket.

5.11 Scalability and Future Plan

The system can be further enhanced to support multiple metro lines, larger station databases, real-time train information, online payment integration, ticket cancellation, 20 ticket history, mobile access, and advanced passenger notifications. The architecture can also be extended to support larger numbers of users and automated metro service operations.

Project Conclusion

The **Metro Ticket Generating System** developed on ServiceNow provides a simple and automated way to book metro tickets and generate digital QR tickets. The system includes station selection, fare calculation, field validation, and QR code generation. ServiceNow automation and notifications help reduce manual work and make the booking process faster and more accurate. The project can be further improved by adding online payments, ticket cancellation, live metro updates, and mobile support. Overall, the project demonstrates how ServiceNow can be used to create an efficient and user-friendly metro ticketing system.